

Foundations of Data Science

Google Advanced Data Analytics Professional Certificate — Course 1 Complete Notes (Updated)

Format: Comprehensive Reference & Exam Prep Guide

Special Integration: RACI Stakeholder Matrix

Module 1: Modern Data Science & Careers

This module outlines the foundational parameters separating traditional analytical processes from modern data science, cataloging standard toolkits, and mapping the trajectory of data disciplines.

1. Data Science vs. Data Analytics

- **Data Analytics:** Concentrates primarily on processing and organizing historical data to identify trends, create diagnostic dashboards, solve defined operational issues, and support active decisions.
- **Data Science:** An expansive framework combining scientific methods, mathematical theory, programming pipelines, and automated algorithms to make highly accurate future predictions and automate real-time decisions.

The Three Vs of Big Data

The expansion of internet infrastructure led to the historical transition into "Big Data" characterized by:

1. **Volume:** Massive data collections requiring distributed systems.
2. **Velocity:** Real-time, continuous data generation and ingestion.
3. **Variety:** Unstructured and semi-structured formats (e.g., streaming logs, text, sensor tracking).

2. The Technical Ecosystem

- **Python:** The primary programming platform for modern statistical engineering. Favored due to broad community support, syntax readability, and seamless machine learning integration.
- **pandas & NumPy:** Python packages specialized for high-performance matrix and tabular operations. The structural foundation of exploratory scripts.
- **SQL (Structured Query Language):** The standard relational database language used to query, join, filter, and extract complex relational datasets.
- **Tableau:** An interactive data visualization platform used to synthesize raw metrics into consumable, dynamic visual narratives.

Module 2: Data-Driven Careers & Professional Ethics

Focuses on enterprise data team structures, distinguishing professional specializations, exploring ethical frameworks, and implementing organizational models like the RACI Matrix.

1. The Professional Matrix

Role Title	Core Focus	Primary Tools	Typical Responsibility
Data Analyst	Diagnostic & descriptive reports; optimization	SQL, Tableau, Excel, Python	Generate monthly performance dashboards.
Data Scientist	Predictive modeling, custom stats models, experiments	Python, SQL, R, Jupyter	Build algorithms predicting user churn behavior.
Data Engineer	Infrastructure pipelines, cloud architectures, database design	SQL, Spark, Cloud SDKs, APIs	Design high-throughput ingestion pipelines.
ML Engineer	Production deployment, deployment infrastructure	Python, TensorFlow, Docker, K8s	Implement recommendation engines at scale.

2. Organizing Teams: The RACI Matrix

The RACI matrix is a critical project management tool used to structure roles, remove operational ambiguity, and define communication flows within complex data workflows:

- **R Responsible (The Doer):** The specific individual or individuals assigned to physically execute the hands-on task and complete the required work.
- **A Accountable (The Owner):** The single person with final decision-making power who is ultimately answerable for the success, failure, and quality of the final deliverable. *Note: To prevent confusion, there should only be exactly one Accountable owner per task.*
- **C Consulted (The Advisor):** Subject matter experts or external stakeholders whose opinions, advice, or feedback are actively sought out before a deliverable can be finalized.
- **I Informed (The Stakeholder):** Team members, executives, or clients who need to be kept updated regarding project milestones and final results but are not directly executing or deciding on the tasks.

3. Data Stewardship & Ethics

- **Data Stewardship:** The ethical and responsible management of the data lifecycle (acquisition, cleaning, validation, protection, and archival).
- **Algorithmic Bias:** Occurs when machine learning systems ingest data containing historic human bias, leading to systemic, unfair predictive results.
- **Privacy Principles:**
 - **Anonymization & Masking:** Scrubbing and isolating Personally Identifiable Information (PII).
 - **Data Minimization:** Limiting collection strictly to the data points necessary for target operations.
 - **Informed Consent:** Explicitly communicating processing intent and securing user approval.

Module 3: Future Trajectory & AI Integration

How data professions adapt to emerging systems, specifically Artificial Intelligence and advanced scripting workflows.

1. Advanced Automated Tools

- **Automated Machine Learning (AutoML):** Automating standard, repetitive modeling sequences (algorithm selection, hyperparameter tuning), enabling analysts to concentrate on framing and business validation.
- **Real-time Stream Analysis:** The rapid shift from batch processing schedules to persistent analytical telemetry.

2. Generative AI integration

Data professionals actively use Large Language Models (LLMs) to write baseline scripts, automate documentation, debug legacy SQL, and translate deep complex outputs into non-technical corporate briefings.

Module 4: The Data Project Workflow (PACE)

The core structural framework utilized to coordinate projects from absolute ambiguity to delivery.

1. The PACE Stages

- **PLAN:** Construct project scope, recognize primary stakeholders, determine technical needs, and author Project Proposals.
- **ANALYZE:** Retrieve, refine, inspect, structure, and visualize raw files. Perform extensive Exploratory Data Analysis (EDA).
- **CONSTRUCT:** Code analytical models, structure features, write predictive pipelines, and balance parameters.
- **EXECUTE:** Deliver insights, write Executive Summaries, present results, and handover production code.

Critical Deliverable: The Project Proposal

Established in the **PLAN** phase. Key contents include the primary business problem, project scope, data requirements, technical constraints, milestones, and measurable success criteria. Aligning expectations early prevents scope creep.

Module 5: Portfolio Projects & Case Scenarios

Applying the PACE structure to distinct client-based simulation environments designed to simulate high-level consulting assignments.

1. Technical Foundation: pandas

An essential library for manipulating structured data. Core commands include:

- `df.head()` / `df.info()`: Display initial records and technical datatypes.
- `df.isnull().sum()`: Compute missing or null inputs across dimensions.
- `df.groupby('category').mean()`: Aggregate metrics by specific classes.
- `df.drop_duplicates(inplace=True)`: Eliminate redundant records.

2. Client Case Scenarios

Scenario A: Automatidata (New York Taxi & Limousine Commission)

- **Business Problem:** TLC wants to improve driver efficiency and passenger experience through rate predictions.
- **Goal:** Optimize NYC taxi ride data to build prediction engines estimating fare totals prior to trips.
- **Data Focus:** Trip times, miles traveled, drop-off locations, payment structures, and fee regulations.

Scenario B: TikTok (Safety & Moderation)

- **Business Problem:** Moderate massive video uploads to prevent toxic or misleading campaigns.
- **Goal:** Differentiate between "claims" (concrete factual statements) and "opinions" (expressive beliefs).
- **Data Focus:** Text sentiment, reporting metrics, user validation status, and interaction rates.

Scenario C: Waze (User Churn Management)

- **Business Problem:** Drive growth and reduce user churn.
- **Goal:** Detect correlation profiles marking users who are prone to abandon the GPS mapping ecosystem.
- **Data Focus:** Weekly driver sessions, alert frequency, system notifications, and drive durations.